

Chapter 1: Introduction to Web Development

Web development involves building and maintaining websites and web applications. It is divided into frontend development, which deals with the user interface, and backend development, which handles server-side logic and databases. Full-stack developers work on both sides. Core technologies include HTML for structure, CSS for styling, and JavaScript for interactivity. Modern web development uses frameworks and tools to streamline the development process.

Chapter 2: HTML and CSS Fundamentals

HTML (HyperText Markup Language) defines the structure of web pages using elements and tags. Semantic HTML improves accessibility and SEO. CSS (Cascading Style Sheets) controls the visual appearance of HTML elements. The box model, flexbox, and CSS grid are essential layout techniques. Responsive design ensures websites look good on all screen sizes using media queries. CSS preprocessors like SASS extend CSS with variables and nesting.

Chapter 3: JavaScript Essentials

JavaScript is the programming language of the web, enabling dynamic and interactive content. It supports variables, functions, arrays, and objects. The DOM (Document Object Model) allows JavaScript to manipulate HTML elements. Event listeners respond to user actions like clicks and keypresses. ES6 introduced modern features like arrow functions, promises, and destructuring. Fetch API enables asynchronous HTTP requests to backend services.

Chapter 4: React Framework

React is a JavaScript library developed by Facebook for building user interfaces. It uses a component-based architecture where UI is broken into reusable pieces. JSX combines HTML and JavaScript syntax. State and props manage data flow between components. React Hooks like `useState` and `useEffect` simplify state management in functional components. The virtual DOM improves performance by minimizing direct DOM manipulation.

Chapter 5: Backend Development with Node.js

Node.js is a JavaScript runtime built on Chrome's V8 engine that enables server-side programming. Express.js is a minimal web framework for building APIs. RESTful APIs use HTTP methods like GET, POST, PUT, and DELETE. MongoDB is a NoSQL database commonly paired with Node.js. Authentication using JWT tokens secures API endpoints. Deployment to cloud platforms like AWS and Heroku makes applications accessible globally.